

School of Arts and Sciences
Department of Computer Science and Information Technology
Fall 2025 – 2026

CSCI250 Introduction to Programming

Class Day :	_____	Class Time: _____	Room: _____
Instructor :	_____		
E-mail :	_____		
Office :	_____	Office Hours: _____	Extension No: _____

I. Course Description

This course introduces the basic concepts and principles of structured programming in Java. It starts with an introduction to Java showing its syntax and the structure of a program in Java then teaches simple data types, control structures, methods, arrays, and strings.

II. Course Objectives

CO-1: The students will be introduced to basic data types and the standard Input and Output.

CO-2: The student will acquire basic principles of algorithmic problem solving, and the skill to write programs in the Java programming language.

CO-3: The student will also know control structures, loops, methods, and arrays.

III. Course Learning Outcomes

Outcome	Details
CLO-1	Describe the concept of variables, their data types, and memory allocation
CLO-2	Demonstrate the ability to solve and implement real-life programs.
CLO-3	Choose the appropriate structure to be used in a specific program.
CLO-4	Observe user-defined methods and employ them to write modular programs.
CLO-5	Use single and multi-dimensional arrays.

V. Mapping to program outcomes (*PLOs are listed at the end of the syllabus*)

Introduction to Programming	PLO -1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9
CLO-1	X								
CLO-2		X							
CLO-3	X		X	X					
CLO-4	X			X					
CLO-5	X		X	X					

VI. Topics Covered

Topic	Details
Writing Programs	Develop Java programs
	Organize program code into modules using methods
Learn Arrays	Understand Arrays
Test and Debug	Test and debug Java programs

VII. Prerequisite – Corequisite

Prerequisite : CSCI205 – Computer Science Overview

VIII. Course Material:

Textbook: Introduction to Java Programming, Daniel Liang, 9th edition

IX. Detailed Schedule

Week		Book Reference	Topic Details
Week 1 Sep 29-03	Lecture 1	Introduction to Java Programming	Read and explain the syllabus Chapter 1: Introduction to Computers, Programs, and Java 1.1 Introduction 1.2 What Is a computer? 1.2.1 Central Processing Unit (Brief) 1.2.3 Memory (Brief) 1.6 The Java language specification, API, JDK, and IDE 1.7 A Simple Java Program 1.10.3 Block Styles
	Lecture 2	Introduction to Java Programming	Chapter 1: Introduction to Computers, Programs, and Java 1.10 Programming Style and documentation [brief] 1.10.1 Appropriate Comments and Comment Styles 1.10.2 Proper Indentation and Spacing 1.11 Programming errors 1.11.1 Syntax Errors 1.11.2 Runtime Errors 1.11.3 Logic Errors <u>It is preferable to show on NetBeans how these errors occur</u>
Week 2 Oct 06-10	Lecture 1	Introduction to Java Programming	Chapter 2: Elementary Programming 2.2 Trace a Program Execution. 2.3 Reading Input from the console (nextDouble() only) Solve Listing 2.2. 2.4 Identifiers 2.5 Variables 2.6 Assignment Statements and Assignment Expressions 2.7 Named Constants. Solve programming exercise 2.2
	Lecture 2	Introduction to Java Programming + Handouts	Chapter 2: Elementary Programming 2.9 Numeric Data Types and Operations 2.9.1 Numeric Types 2.9.2 Numeric Operators (+, -, /, *, %) focus on the remainder and integer division. 2.10 Numeric Literals 2.11 Evaluating Expressions and Operator Precedence 2.3 Reading Input from the Console (Cont.) Solve Programming exercise 2.6 Exercise: Write a program that asks the user to enter the time in milliseconds. Then the program extracts the hours, minutes, and seconds (using / and %)
Week 3 Oct 13-17	Lecture 1	Introduction to Java Programming + Handouts	Chapter 2: Elementary Programming 2.13 Augmented Assignment Operators 2.14 Increment and Decrement Operators 2.15 Numeric Type Conversions + Division of two integers yields an integer + Decimal places (keeping 2 digits after the decimal point). Solve programming exercise 2.16 (Math.sqrt included)
	Lecture 2	Introduction to Java Programming + Handouts	Chapter 2: Elementary Programming 2.17 Character Data Type and Operations 2.17.2 Escape Characters (\n and \t only) 2.17.3 Casting between char and Numeric Types (char and int only) 2.3 Reading Char Input from the Console (next().charAt(0)) 2.18 The String type (concatenation, next() and nextLine()) Handouts: ASCII table Exercise: A shop is offering discounts. Write a program that asks the user to enter his name and the product brand. The user also enters the initial price and the discount. The program should compute and display the new price after the discount

Week 4 Oct 20-25	Lecture 1	Introduction to Java Programming	Chapter 3: Selections 3.1 Introduction 3.2 Boolean Data Type 3.3 if Statements 3.2 Solve Listing 3.2 3.5 Two-Way if-else Statements QUIZ 1 – Chapters 1 & 2
	Lecture 2	Introduction to Java Programming + Handouts	Chapter 3: Selections 3.6 Nested if and multi-Way if-else Statements 3.7 Common Errors in Selection Statements. Solve letter grade exercise Figure 3.4 page 91.
Week 5 Oct 27-31	Lecture 1	Introduction to Java Programming	Assignment 1 Chapter 3: Selections 3.11 Logical Operators + Examples Solve Listing 3.7
	Lecture 2	Introduction to Java Programming	Chapter 3: Selections 3.15 The Switch Case Solve programming exercises 3.8, and 3.12 Solve listing 3.10
Week 6 Nov 03-07	Lecture 1	Introduction to Java Programming	Exam 1 – Chapters 1,2 and 3
	Lecture 2	Introduction to Java Programming	Chapter: 4 Loops 4.1 Introduction 4.2 The while Loop 4.2.2 Loop Design Strategies 4.2.4. Controlling a Loop with a sentinel Value 4.3 The do-while Loop. Solve programming exercise 4.8, 4.16
Week 7 Nov 10-15	Lectures 1 & 2	Introduction to Java Programming + Handouts	Chapter: 4 Loops 4.4 The for Loop 4.5 Which Loop to Use? 4.8.1. Problem: Finding the Greatest Common Divisor 4.9. Keyword break. Solve programming exercise 4.14(use break) Handouts: Explain Math.random(), + exercises on Math.random()) Write programs to calculate the below sum1 = 1 + 2 + ... + n sum2 = 1 + 1/2 + 1/4 + ... + 1/n factorial = 1 * 2 * ... * n Write a program to display whether an integer is prime or not
Week 8 Nov 17-21	Lectures 1 & 2	Introduction to Java Programming	Assignment 2 QUIZ 2 – Chapter 3 (Excluding Nested Loops) <i>Continue week 7 if needed</i> Chapter: 4 Loops 4 6 Nested Loops. Solve programming exercises 4.10, 4.20 and 4.18 (A and B) 4.24, and 4.33
Week 9 Nov 24-28	Lecture 1 & 2	Introduction to Java Programming	Chapter 5: Methods 5.1 Introduction 5.2 Defining a Method 5.3 Calling a Method. Solve Listing 5.1. 5.4 void Method Example Solve Listing 5.3 5.5 Passing Parameters 5.9 Scope of Local Variables Solve programming exercises 5.2, and 5.26

Week 10 Dec 01-05	Lectures 1 & 2	Introduction to Java Programming	QUIZ 3 – Chapter 5 (Excluding 5.10) Chapter 5: Methods 5.10 The Math Class 5.10.2. Exponent Methods (sqrt only) Solve programming exercises 5.17 and 5.18
Week 11 Dec 08-12	Lecture 1	Introduction to Java Programming	Chapter 6 : Single-Dimensional Arrays 6.1 Introduction 6.2 Array Basics 6.2.1 Declaring Array Variables 6.2.2 Creating Arrays 6.2.3 Array Size and Default Values 6.2.4 Array Indexed Variables 6.2.5 Array Initializers 6.2.6 Processing Arrays
	Lecture 2	Handouts	Chapter 6: Single-Dimensional Arrays Practice Exercises: a) Create an array of integers named List. Use the shorthand notation to fill it with ten values. b) Write the code to sum up all the element in the array. c) Write the code to find the highest value in the array. d) Sum of all even numbers in an array. e) Sum all elements located at an even index.
Week 12 Dec 15 – 19	Lectures 1 & 2	Introduction to Java Programming + Handouts	Chapter 6: Single-Dimensional Arrays 6.5. Copying Arrays 6.10.1. The linear Search Approach 6.6 Passing Arrays to Methods 6.6.1 Passing Array Arguments (Explain the difference between passing by value and by reference) 6.7 Returning an Array from a Method + Practice Exercises
Week 13 Dec 22-23	Lecture 1	Introduction to Java Programming	Wed and Thu – Christmas Holiday Chapter 6: Single-Dimensional Arrays Solve programming exercise: 6.2, 6.4, 6.10 and 6.26
Week 14			Assignment 3 Christmas and New Year Holiday
Week 15 Jan 04-08	Lecture 1 & 2	Introduction to Java Programming	Tue - Armenian Christmas Holiday Chapter 7: Multidimensional Arrays 7.1 Introduction 7.2 Two-Dimensional Array Basics 7.2.1 Declaring Variables of 2-D Arrays and Creating 2- D Arrays. 7.2.2 Obtaining the Lengths of 2-D Arrays 7.3 Processing 2-D Arrays: - Initializing arrays with input values - Initializing arrays with random values Processing 2-D Arrays practice exercises: Printing Arrays - Summing all elements - Summing elements by column - Find the largest sum in a row

X. Course Assessment

Activity	Percentage
Quizzes	10%
Exam 1	20%
Midterm	35%
Final Exam	35%

XI. Additional Notes/Instructions

Grading of assignments

- (20%) Correctness -- does the program run correctly?
- (20%) Style -- does the code adhere to class documentation standards? Is the code indented properly? Are the variable names mnemonic?
- (20%) Design -- is the program adequately decomposed (e.g., are the functions small enough to be comprehensible)?
- (40%) Completeness and Logic -- is the functionality of the program complete?

Instructions for Exams

- Time allowed: **60 minutes** whereas **75 minutes** for the Midterm and **120 minutes** for Final Exam.
- **Cheating** in any way, form or method will be rewarded with an **F grade**.
- You should read each question carefully before answering.
- You should solve the questions that you are confident about their answers first.
- **Calculators not allowed**
- **Neither an Open Book nor an Open Notes exam.**
- **Laptop and USB not allowed**

XII. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam’s grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

LIU E-mails

LIU students should use their LIU e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

University Peer Review Policy

This is a new policy at LIU in which during the semester, all classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4	D+	72	2.2	B+
65	1.5		73	2.3	
66	1.6		74	2.4	
75	2.5	C+	83	3.3	
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9	B	87	3.7	
80	3		88	3.8	A
81	3.1		89	3.9	
82	3.2		>=90	4	

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

Computer Science Outcomes

Graduates of the Computer Science Program will have the ability to:

	Program Outcomes
category -1 Underlying Conceptual Basis for Informatics	PLO -1 Describe and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study
	<i>PLO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>
category -2 Analysis	PLO -2 Select and use relevant analytic, modelling and simulation methods
	PLO-3 Choose appropriate solution patterns, algorithms and data structures
category -3 Design and Implementation	PLO -4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PLO-4 for CSIT=>Specify and design computing/network hardware/software which meet specified requirements</i>
category -4 Economic, legal, social, ethical and environmental context	PLO -5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems
category -5 Informatics practice	PLO -6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PLO -6 for CSIT=> Demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
category -6 Other Professional Skills and Competences	PLO -7 Communicate effectively both verbally and using a variety of communications media
	PLO -8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PLO -9 Participate effectively in informatics group-working